COMP 3311: Database Management Systems

Lecture 21 Exercises
Query Processing: Size Estimates; Query Optimization

```
select sName
from Sailor
where rating=7
  and age=40;
```

Exercise 1: Given the relation Sailor(<u>sailorId</u>, sName, hkid, rating, age) and the query:

n_{Sailor} = 10,000 B_{Sailor} = 1,000 pages $bf_{Sailor} = \lceil 10{,}000 / 1{,}000 \rceil = 10$

V(rating, Sailor) = 10 (10 distinct rating values)

V(age, Sailor) = 100 (100 distinct age values)

SC(rating=7, Sailor) = n_{Sailor} / V(rating=7, Sailor) = 10,000 / 10 = 1,000 tuples

SC(age=40, Sailor) = n_{Sailor} / V(age=40, Sailor) = 10,000 / 100 = 100 tuples

Estimate the page I/O cost to process the query using the following alternative plans and assuming uniform distribution and attribute independence. *Ignore the cost of searching any indexes.*

a) file scan page I/O cost

b) binary search page I/O cost
   i. Search a file ordered on rating

   ii. Search a file ordered on age

c) single B^+-tree index (on either attribute) page I/O cost
   i. index on rating only

   ii. index on age only

d) multiple B^+-tree indexes (on both rating and age)

Exercise 2: Given the following relations and the information about them, estimate the *minimum page I/O cost* to process the query shown on the right *using materialization* and the join order

(Sailor JOIN $\sigma_{rDate='01-JAN-2025'}$ Reserves) JOIN $\sigma_{color='red'}$Boat

```
select *
from Sailor natural join Reserves natural join Boat
where rDate='01-JAN-2025'
  and color='red';
```

Sailor(<u>sailorId</u>: 2 bytes, sName: 10 bytes, hkid: 10 bytes, rating: 2 bytes, age: 2 bytes)

Reserves(<u>*sailorId*</u>: 2 bytes, <u>*boatId*</u>: 2 bytes, <u>rDate</u>: 7 bytes) Boat(<u>*boatId*</u>: 2 bytes, bName: 10 bytes, color: 10 bytes)

- There are 10,000 Sailor tuples, 100,000 Reserves tuples and 1,000 Boat tuples.

$bf_{Sailor} = 10$	$bf_{Reserves} = 23$	$bf_{Boat} = 8$	Page size = 260 bytes
B_{Sailor} = 1,000 pages	$B_{Reserves}$ = 4,348 pages	B_{Boat} = 125 pages	M = 100 pages

- There are the following indexes:
  – hash index on sailorId for Sailor (no overflow buckets).
  – hash index on boatId for Boat (no overflow buckets).
  – clustering B^+-tree on rDate for Reserves (2 levels).

<u>Some useful statistics:</u>
- Reserves has 1,000 unique rDates.
- 10% of boats are red.
- A sailor has on average 10 reservations.

For each operation, give only the strategy and cost calculation that results in the <u>overall</u> minimum page I/Os.

C_1: Cost of computing $TempC_1$ = (Sailor JOIN $\sigma_{rDate='01-JAN-2025'}$Reserves)

Best strategy: ____________________

Page I/O cost calculation for best strategy for Sailor JOIN $\sigma_{rDate='01-JAN-2025'}$Reserves:

Page I/O cost to write $TempC_1$: __________

Page I/O cost for best C_1 strategy that minimizes overall page I/O cost: ____________

C_2: Cost of computing $TempC_2$ = $\sigma_{color='red'}$Boat (no index on color)

Best strategy: ____________________

Page I/O cost calculation for best strategy for $\sigma_{color='red'}$Boat:

Page I/O cost to write $TempC_2$: __________

Page I/O cost for best C_2 strategy that minimizes overall page I/O cost: ____________

C_3: Cost of $TempC_1$ JOIN $TempC_2$

Best strategy: ____________________

Page I/O cost calculation for best strategy for $TempC_1$ JOIN $TempC_2$:

Page I/O cost for best C_3 strategy that minimizes overall page I/O cost: ____________

<u>Minimum query processing page I/O cost</u>: ____________________

Name: (1) ______________________/________________ **Student#:** (1) __________________ **Date:** ___________
Family/Given (PRINT) Given/First (PRINT)

Name: (2) ______________________/________________ **Student#:** (2) __________________
Family/Given (PRINT) Given/First (PRINT)

This is a partner exercise. YOU MUST DO THIS EXERCISE WITH A PARTNER. Individual submissions will not be graded.
A genuine effort must be made to complete this exercise to obtain full marks.
SUBMIT ONLY ONE COMPLETED PARTNER WORKSHEET.

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Exercise 3: Using the information given for Exercise 2, estimate the *minimum page I/O cost* to process the query using ***materialization***, the join order Sailor JOIN ($\sigma_{\text{rDate='01-JAN-2025'}}$ Reserves JOIN $\sigma_{\text{color='red'}}$Boat) and the given strategies.

C_1: Cost of computing $\text{TempC}_1 = \sigma_{\text{rDate='01-JAN-2025'}}\text{Reserves}$

Strategy $C_1$1: linear search on Reserves

Page I/O cost for selection: ____________________

Page I/O cost to write TempC_1: ____________________

Total page I/O cost for Strategy $C_1$1: ______________________

Strategy $C_1$2: index lookup using B^+-tree index on Reserves.rDate

Page I/O cost for selection: ____________________

Page I/O cost to write TempC_1: ____________________

Total page I/O cost for Strategy $C_1$2: ______________________

C_2: Cost of computing $TempC_2 = TempC_1$ JOIN $\sigma_{color='red'}$Boat (no index on color)

Strategy $C_2$1: block nested-loop join

Page I/O cost to read $TempC_1$: ____________________

Page I/O cost to read Boat: ____________________

Page I/O cost to write $TempC_2$: ______________________

Total page I/O cost for Strategy $C_2$1: ______________________

Strategy $C_2$2: indexed nested-loop join using hash index on Boat.boatId

Page I/O cost to read $TempC_1$: ____________________

Page I/O cost to use hash index on Boat: ____________________

Page I/O cost to write $TempC_2$: ______________________

Total page I/O cost for Strategy $C_2$2: ______________________

C_3: Cost of Sailor JOIN $TempC_2$ using indexed nested loop (hash index on Sailor.sailorId)

Strategy $C_3$1: block nested-loop join

Page I/O cost to read $TempC_2$: ______________________

Page I/O cost to read Sailor: ______________________

Total page I/O cost for Strategy $C_3$1: ______________________

Strategy $C_3$2: indexed nested-loop join using hash index Sailor.sailorId

Page I/O cost to read $TempC_2$: ______________________

Page I/O cost to use hash index on Sailor: ______________________

Total page I/O cost for Strategy $C_3$2: ______________________

Minimum query processing page I/O cost:

____

Submit this exercise worksheet to the instructor by **3:50 p.m. today.**

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Exercise 4: Employee(empId: 4 bytes, name: 35 bytes, title: 2 bytes, salary: 5 bytes, deptId: 4 bytes)
Department(deptId: 4 bytes, deptName: 25 bytes, location: 7 bytes)
DeptProject(deptId: 4 bytes, projectId: 4 bytes)
Project(projectId: 4 bytes, projectTitle: 20 bytes, budget: 6 bytes, report: 970 bytes)

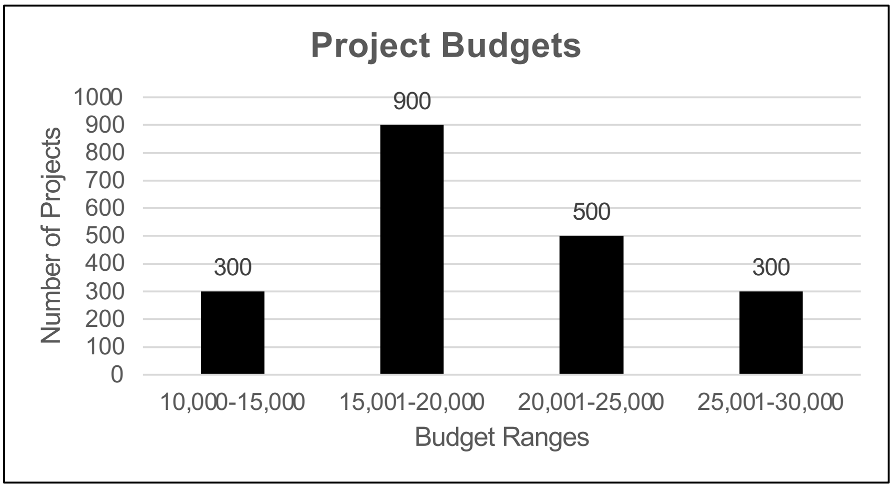
Employee:	50 bytes/tuple;	20,000 tuples	250 pages
Department:	36 bytes/tuple;	100 tuples	1 page
DeptProject	8 bytes/tuple	4,000 tuples	8 pages
Project:	1,000 bytes/tuple;	2,000 tuples	500 pages

All foreign keys are not null. Page size: 4,000 bytes Buffer pages: 12

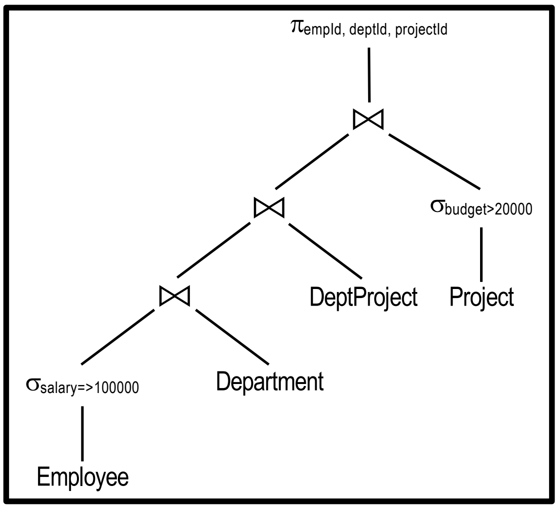
Employee salaries: uniformly distributed in the range 10,000 to 110,000.

Project budgets: distributed in the range 10,000 to 30,000 according to the histogram below.

- There is a clustering B^+-tree index with 3 levels on salary for Employee.
- There is a hash index on deptId for Department; Department is ordered on deptId.
- There is a hash index on projectId for Project; Project is ordered on projectId.




```
select distinct empId, deptId, projectId
from Employee natural join Department
     natural join DeptProject
     natural join Project
where salary=>100000
  and budget>20000;
```



Name: (1) ______________________/________________ **Student#:** (1) __________________ **Date:** ___________

Family/Given (PRINT) Given/First (PRINT)

Name: (2) ______________________/________________ **Student#:** (2) __________________

Family/Given (PRINT) Given/First (PRINT)

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a) Estimate the result size of the query in tuples and in pages for the given relational algebra query evaluation tree. ***Explain clearly your size estimates.***

Estimated result size (tuples): ______________________ Estimated result size (pages): ________

b) Estimate the query processing page I/O cost using the relational algebra query evaluation tree and the steps given below. The goal is to minimize the total number of page I/Os needed to process the query. Reorder operations, as necessary, to reduce the page I/O cost. Where possible, use pipelining rather than materialization (i.e., keep intermediate results in the buffer where possible). Assume the file organizations and indexes described above. For each step keep only the attributes necessary to process the query.

Step 1: $\sigma_{salary=>100000}$Employee $\Rightarrow$ Result A

Strategy: ____

Page I/O cost calculation and explanation:

Step 1 minimum page I/O cost: ______________________ Step 1 result size (pages): ___________

Step 2: Result A $\bowtie$ Department $\Rightarrow$ Result B

Strategy: ____

Page I/O cost calculation and explanation:

Step 2 minimum page I/O cost: ______________________ Step 2 result size (pages): ___________

Step 3: Result B ⋈ DeptProject ⟹ Result C

Strategy: ____

Page I/O cost calculation and explanation:

Step 3 minimum page I/O cost: ____________________ Step 3 result size (pages): ________

Step 4: $\sigma_{budget>20000}$Project ⟹ Result D

Strategy: ____

Page I/O cost calculation and explanation:

Step 4 minimum page I/O cost: ____________________ Step 4 result size (pages): ________

Step 5: Result C ⋈ Result D

Strategy: ____

Page I/O cost calculation and explanation:

Step 5 minimum page I/O cost: ____________________ Step 5 result size (pages): ________

Query processing page I/O cost: ____________________ Query result size (pages): ________